

Algorithmics	Student information	Date	Number of session
	UO: 300737	14/04/2025	7
	Surname: Hoyos		
	Name: Ignacio		



PART A: Branching Heuristic

We are going to follow a branching heuristic aiming for the best available node, that will be the one that give us a current cost nearest to zero.

PART B: Measurements

NULLPATHBB (times in milliseconds WITHOUT OPTIMIZATION)

n	tAcumulated/100
5	2.21
10	326.14
15	5 195
20	OoT
25	OoT
30	OoT

According to the code it must have a factorial complexity, and looking at the incredible fast increase in the taken time, it seems so. If we compare these results with the backtracking ones... On the one hand, both have the “same” complexity $O(n!)$ but on the other hand, we can see that our improvements to discard not valid paths have make a time improvement. If we approximate our complexities to a exponential ones, Backtracking seems n^{11} while Branch and Bounds looks n^7 .

$$T(n)=c \cdot n^k \rightarrow k = \frac{\log(T2)-\log(T1)}{\log(n2)-\log(n1)}$$

$$\text{Backtracking} \rightarrow \frac{\log(130.44)-\log(0.06)}{\log(10)-\log(5)} = 11$$

$$\text{Branch and Bounds} \rightarrow \frac{\log(326.14)-\log(2.21)}{\log(10)-\log(5)} = 7$$